Project Progress Report 2 – Model 1 (Final Regression Model)

Port Authority Tunnels & Bridges – Traffic Analytics

1. Model Objective

The objective of Model 1 is to predict monthly TOTAL_TRAFFIC using seasonality, facility characteristics, mobility metrics, and regional transit demand indicators. The goal is to identify structural drivers of roadway usage and develop a forecasting framework to support revenue planning, congestion pricing analysis, and operational capacity decisions.

2. AutoML Model Development (H2O)

2.1 Dependent Variable

TOTAL_TRAFFIC (Monthly Vehicle Volume)

2.2 Independent Variables (No-Leakage Configuration)

YEAR, MONTH, Facility, Direction, Avg_Speed, Freeflow, Delta, PABT_Bus_Volume, PABT_Passenger_Volume.

The following variables were intentionally excluded to prevent structural leakage: TOTAL_EZPASS, TOTAL_CASH, TOTAL_VIOLATIONS.

2.3 Selected Algorithm

Gradient Boosting Machine (GBM) selected via H2O AutoML based on lowest cross-validated RMSE.

2.4 Hyperparameters Used

ntrees=232, max_depth=8, sample_rate=0.8, col_sample_rate=0.8, 5-fold cross-validation (Modulo assignment), distribution=gaussian, stopping_metric=deviance, stopping_tolerance=0.05.

2.5 Model Performance (Cross-Validation)

RMSE: 18,111.80
MAE: 11,408.24
R^2: 0.999168
Observations: 146

Given that average monthly traffic is approximately 10 million vehicles, this corresponds to approximately 0.18% relative error, indicating extremely high predictive accuracy.

2.6 Variable Importance

Top Drivers Identified:
1. MONTH (61.19%) – Strong seasonal demand patterns.
2. PABT_Passenger_Volume (22.00%) – Regional travel demand indicator.

3. PABT_Bus_Volume (9.30%) – Transit-roadway interaction effects.
4. YEAR (7.32%) – Structural growth trend.
Remaining variables (Facility, Direction, Speed metrics) contributed marginal incremental variation.

2.7 Business Interpretation

The model demonstrates that traffic demand is structurally driven by seasonality and broader regional mobility patterns. Transit passenger volumes act as leading indicators of roadway usage, highlighting strong interdependence between transit and tunnel/bridge demand. Mobility speed metrics appear more reflective of congestion outcomes than primary demand drivers.

The model provides a reliable framework for:
- Revenue forecasting
- Capacity planning
- Congestion pricing simulations
- Infrastructure optimization decisions

3. Python Model Replication

The no-leakage GBM model was replicated in Python using sklearn's GradientBoostingRegressor with 5-fold cross-validation and aligned hyperparameters.

3.1 Python No-Leakage Results

Cross-Validated RMSE: 315,453
Relative Error: 3.15%

Differences between H2O and Python RMSE values are attributable to implementation differences, cross-validation mechanics, and categorical encoding strategies. Both models confirm strong predictive structure in the data.

4. Final Conclusion

Model 1 successfully identifies and quantifies the structural drivers of Port Authority traffic demand. Seasonality and regional transit volumes dominate traffic variation, while payment variables are not required for accurate forecasting. The final no-leakage GBM model provides a robust, interpretable, and operationally useful forecasting tool suitable for advanced planning and policy evaluation.